

Universidad Tecnológica Nacional - Facultad Regional Buenos Aires

Ingeniería Industrial

Mecánica y Mecanismos

Trabajo Práctico N°2: Dinámica del Punto

Grupo N° 3

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Curso: I4053

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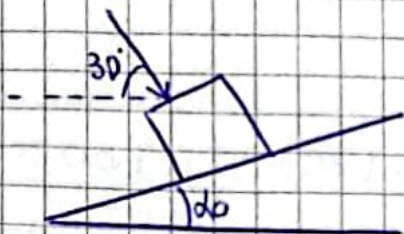
Presentación	Fecha	Devolución
1°	30/06/2023	
2°		
3°		
4°		
Aprobación:		
		Firma

① $m = 45 \text{ kg}$ em repouso

$P = f \text{ constante}$

$\mu = 0,35$

$v = 0,6 \text{ m/s} \longrightarrow x = 0,9 \text{ m}$ para arriba



Tenendo em conta que

$$a = \frac{dv}{dt}$$

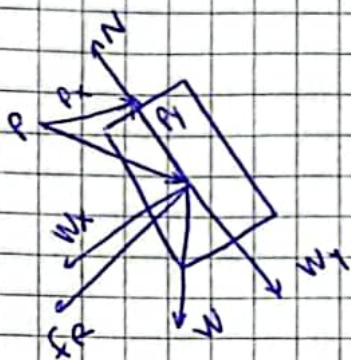
$$v = \frac{dx}{dt}$$

$$dl = \frac{dx}{v}$$

$$a = \frac{v dv}{dx} \longrightarrow a dx = v dv$$

$$a \int_0^{0,9} dx = \int_0^{0,6} v dv$$

$$0,9a = 0,6^2/2 \longrightarrow a = 0,2 \text{ m/s}^2$$



$$\Sigma F_y = -P \sin 50 + mg \cos 20 = ma_y$$

$$P \sin 50 = N mg \cos 20$$

$$mg \cos 20 + P \sin 50 = N$$

$$\Sigma F_x = P \cos 50 - mg \sin 20 - \mu N$$
$$= m a_x$$

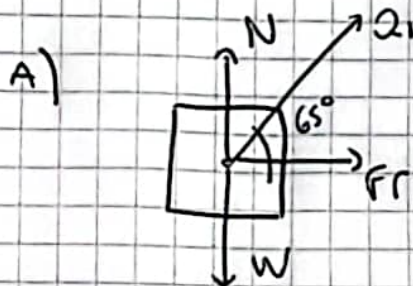
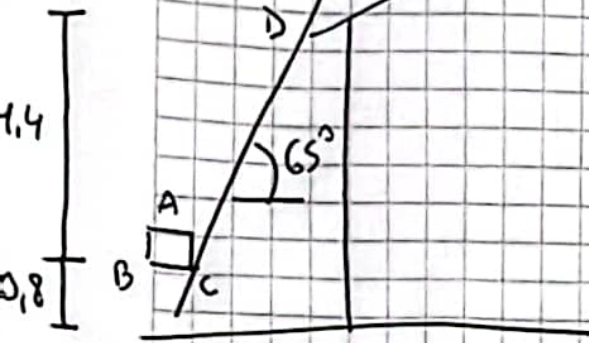
$$P \cos 50 - mg \sin 20 - \mu (mg \cos 20 + P \sin 50) = 0,2 m$$

$$P(-\mu \sin 50 + \cos 50) - \mu mg \cos 20 - mg \sin 20 = 0,2 m$$

$$P = \frac{m(a_x + \mu g \cos 20 + g \sin 20)}{-\mu \sin 50 + \cos 50} \approx 820 \text{ N}$$

②

$$\mu = 0,3$$



$$\sum F_y \rightarrow N - W = m \cdot a_1 \sin 65 \quad (1)$$

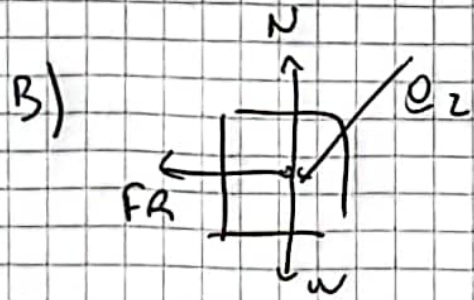
$$\sum F_x \rightarrow F_r = m \cdot a_1 \cos 65$$

$$\mu N = m \cdot a_1 \cos 65 \quad (2)$$

$$\mu [m(a_1 \sin 65 + g)] = m \cdot a_1 \cos 65$$

$$\mu \cdot g = a_1 [\cos 65 - \mu \sin 65]$$

$$\boxed{a_1 = 19,9 \text{ m/s}^2}$$



$$\sum F_x \rightarrow N = a_2 \cos 65 \cdot m$$

$$\sum F_y \rightarrow N - W = -m \cdot a_2 \sin 65$$

$$a_2 \cos 65 \frac{m}{m} - m \cdot g = -m a_2 \sin 65$$

$$\boxed{a_2 = 7,3 \text{ m/s}^2}$$

3)

$$m = 60 \text{ kg}$$

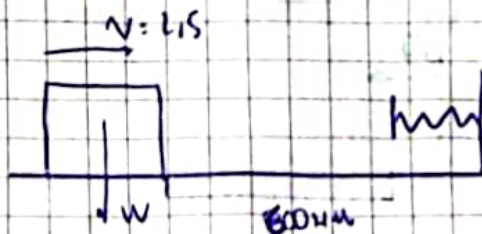
$$v_0 = 2,5 \text{ m/s}$$

$$k = 20 \text{ kN/m}$$

$$g = 10 \text{ m/s}^2$$

comprimido 120 mm

comp max = 40 mm



$$T_{1-2} = F dx$$

$$F_r = \mu N$$

$$T_{1-2} = -F_r dx = -\mu N dx$$

$$-\mu mg (\cancel{0} + 0,04) \text{ m} = \frac{T_{1-2}}{F_r}$$

a) P(1)

$$E_{c1} = \frac{1}{2} m v^2 = 187,5 \text{ J}$$

P(2) freio $E_c = 0$

$$T_{el} = \frac{1}{2} k \Delta x^2$$

$$T_{el} = \frac{1}{2} k (0,16^2 - 0,12^2) = -112 \text{ J}$$

$$D_{em} = T_{1-2}$$

$$E_{c2} - E_{c1} = T_{1-2} \rightarrow -\mu g m (0,64) \text{ m} - 112 = -E_{c1}$$

$$\mu = 0,197$$

b) $T_{2-3} = -0,197 m g \cdot 0,64 = -75,5 \text{ J}$

$$T_{el} = \frac{1}{2} k (0,16^2 - 0,12^2) = +112 \text{ J}$$

$$D_{em} = T_{2-3} \rightarrow \frac{1}{2} m v_f^2 = (112 - 75,5) \text{ J}$$

$$v_f = 1,103 \text{ m/s}$$

4)

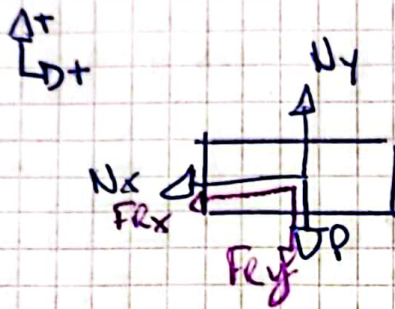
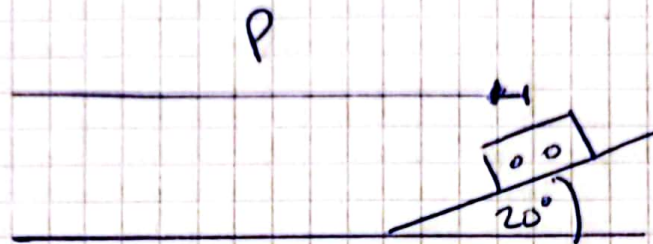
$$m = 1700 \text{ g}$$

$$\alpha = 20^\circ$$

$$P = 100 \text{ m}$$

$$\mu = 0.2$$

$$V_{\text{CIE}} = m \dot{x}^2?$$



$$V_{\text{MAX}} = \sqrt{2mP}$$

$$V_{\text{MAX}} = \sqrt{6 \frac{\text{m}}{\text{s}^2} - 100 \text{ m}}$$

$$\boxed{V_{\text{MAX}} = 24,5 \frac{\text{m}}{\text{s}}}$$

$$\Sigma F_x = N_x + F_{rx} = m \cdot a_m$$

$$\textcircled{1} N \cdot \sin \alpha + F_r \cos \alpha = m a_m$$

$$\textcircled{1} N (\sin 20 + \mu \cos 20) = m a_m$$

$$\Sigma F_y = N_y - P - F_r \sin 20 = 0$$

$$N \cos 20 = P + \mu N \sin 20$$

$$N = (\cos 20 - \mu \sin 20) = P \rightarrow \textcircled{2} N = \frac{P}{\cos 20 - \mu \sin 20}$$

$$\textcircled{2} \text{ EN } \textcircled{1}$$

$$\frac{m g}{\cos 20 - \mu \sin 20} \cdot (\sin 20 + \mu \cos 20) = m a$$

$$\boxed{a_m = 6 \frac{\text{m}}{\text{s}^2}}$$

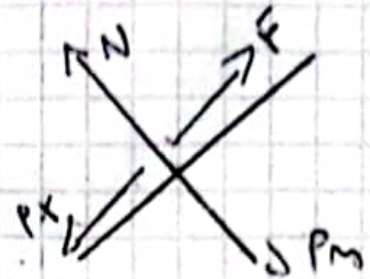
$$5) F = 400 \text{ kg} \rightarrow m = 40,81 \text{ kg}$$

$$F = 320 \text{ t}^2$$

$$v_i = 2 \text{ m/s} \rightarrow T = 0$$

$$T_0 \alpha = \frac{2}{10}$$

$$\alpha = 28^\circ$$



$$\sum F_x = F - P_x = m \cdot a_x = m \cdot \frac{dv}{dt}$$

$$320 \text{ t}^2 - 400 \sin 28^\circ = 40,81 \cdot \frac{dv}{dt}$$

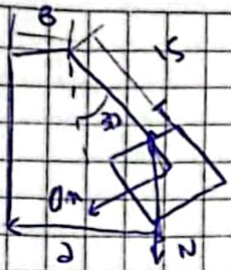
$$(320 \text{ t}^2 - 187,8) dt = 40,81 \cdot dv$$

$$320 \frac{t^3}{3} - 187,8 t \Big|_0^2 = 40,81 \cdot v \Big|_{v_0}^{v_F}$$

$$477,7 = 40,81 \cdot v_F - 81,62$$

$$\boxed{13,68 \frac{\text{m}}{\text{s}} = v_F}$$

6)



$$m = 550 \text{ kg}$$

$$a = 8 + 15 \sin(30) = 15,5$$

$$\sum F_x = -t \sin(30) = -m a_m \quad (1)$$

$$\sum F_y = -W + t \cos(30) = 0 \quad (2)$$

donde $t = \frac{m g}{\cos(30)}$

reemplazo en (1)

$$\frac{-m g}{\cos(30)} \sin 30 = -m a_m$$

$$g \tan 30 = a_m \quad \text{---} \quad a_m = v^2 / a$$

$$v = \sqrt{a_m a}$$

$$v = \sqrt{g \tan 30 a} = \sqrt{9,8 \tan 30 15,5}$$

$$v = 9,13 \text{ m/s}$$

$$\sum F_x = -N \sin 30 = -m a_m \quad -60 \text{ kg}$$

$$\sum F_y = -W + N \cos 30 = 0$$

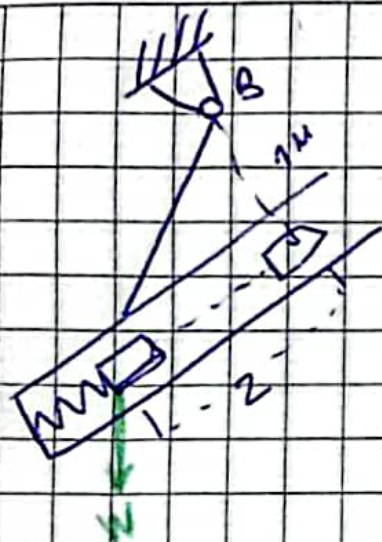
$$N_y = N \cos 30 = m g$$

$$N_y = 60 \cdot 9,8 \rightarrow N_y = 588$$

$$N_x = (-N \sin 30) = -60 \cdot 9,8 \cdot \tan 30 \rightarrow N_x = 339,5$$

$$\therefore N = \frac{N_x}{\sin 30} = 678,9$$

4



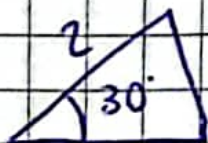
$$30 \text{ kg} = 300 \text{ N}$$

$$k = 5 \frac{\text{kg}}{\text{m}} = 50 \text{ N/m}$$

$$m = 10 \text{ kg}$$

$$\mu_1 = 0,5 ; \mu_2 = 0$$

$$E_{ci} = E_{cf} - T_{\Delta c}$$



$$|AB| = \sqrt{5} \text{ m}$$

$$|BC| = 1 \text{ m}$$

$$T_w = -m g 2 \sin(30) = -100 \text{ J}$$

$$T_{fc} = -1/2 m [(2,5u)^2 - (0,5u)^2] = -150 \text{ J}$$

$$T_f = f \cdot (AB - BC) = 300 \text{ N} \cdot (\sqrt{5} \text{ m} - 1 \text{ m}) = 370,82 \text{ J}$$

$$-100 \text{ J} - 150 \text{ J} + 370,82 \text{ J} = 1/2 m v_c^2$$

$$v_c = 4,92 \text{ m/s}$$

8

$$m_A = 50 \text{ TN}$$

$$v_{0A} = 2 \text{ Km/h}$$

$$\tau = 3 \text{ up}$$

$$m_B = 60 \text{ TN}$$

$$v_{0B} = 0 \text{ Km/h}$$

$$a) v_F = ?$$

$$b) f_{imp} = ?$$

$$m_A \cdot v_{0A} + m_B \cdot v_{0B} = (m_A + m_B) \cdot v_F$$

$$v_F = \frac{40000 \text{ kg} \cdot 0,55 \text{ m/s}}{100000 \text{ kg}}$$

$$v_F = 0,22 \text{ m/s}$$

Vapor A

$$\Delta p_A = m_A (v_{FA} - v_{0A}) = 40000 \text{ kg} (0,22 \text{ m/s} - 0,55 \text{ m/s})$$

$$I = -13200 \text{ Kg m/s}$$

$$\Delta p_A = f_{med, t} \rightarrow f_{med, A} = -4400 \text{ N}$$

Vapor B

$$\Delta p_B = m_B (v_{FB} - v_{0B}) = 60000 \text{ kg} (0,22 \text{ m/s} - 0)$$

$$I = 13200 \text{ Kg m/s}$$

$$f_{med, B} = 4400 \text{ N}$$